Scaling Laws for Neural Language Models (Kaplan et al. 2020)

Cross-entropy loss in autoregressive Transformer language models follows smooth power-law relationships with parameter count, dataset size, and compute budget over multiple orders of magnitude.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Language model performance improves predictably as a power-law function of model parameters N, dataset size D, and total training compute C, with smooth scaling spanning more than six orders of magnitude.

EVIDENCE AND CAUSAL ASSESSMENT

Systematic multi-scale experiments varying N, D, and C independently isolate the causal trade-offs governing cross-entropy test loss, showing that architectural hyper-parameters have minimal impact compared to scale.

EFFECT SIZE AND PERSISTENCE

Test loss scales as L(N) \propto N^{-lpha_N}, L(D) \propto D^{-lpha_D}, and L(C) \propto C^{-lpha_C}, allowing researchers to accurately forecast the loss of billion-parameter models from small-scale training runs.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%). Although Hoffmann et al. (2022) adjusted Kaplan's compute-optimal token-to-parameter ratio from $1:1to roughly $20:1, the fundamental finding of smooth power-law scaling remains robust.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This does not mean that scaling parameters alone solves truthfulness, safety, or factuality; lower cross-entropy loss merely indicates more accurate token probability modeling.

WHAT ACTION IS REASONABLY JUSTIFIED

Use small-scale scaling law projections to optimize compute and data allocation before committing to multi-million-dollar training runs.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., ... & Amodei, D. (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361.

- Hoffmann, J., et al. (2022). Training Compute-Optimal Large Language Models. arXiv:2203.15556.